

Yixiao Chen

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EXPERIENCE

Perception Engineer, Motional

07/2026 – Present

Boston, MA

- Contributing to Motional's mission to make driverless vehicles a safe, reliable, and accessible reality.

Research Assistant, Northeastern University

05/2024 – 06/2026

Boston, MA

- Pruned and quantized Mixture-of-Experts (MoE) LLMs (e.g., DeepSeek V3) with MoE pruning algorithms and llama.cpp, and deployed them on edge platforms, including AMD Windows systems with Vulkan and HIP/ROCm backends.
- Built a real-time conversational system integrating LLMs and Text-to-Speech (TTS) for seamless, natural human-computer interaction.
- Deployed digital avatar models with CoreML and TensorRT, accelerating inference to real-time performance.

Machine Learning Engineer, EmbodyX Inc. & Albao LLC

05/2024 – 06/2026

Boston, MA (Remote)

- Accelerated and deployed Vision-Language-Action (VLA) and World Action Models (WAM) for embodied AI.
- Worked across robotic arm (UR) and humanoid (Unitree, AGIBOT) platforms.
- Developed a trajectory prediction assistant with transformer-based models for driver's license learning exams.
- Built video data filtering with face centering and scene detection, and audio filtering with multi-expert mean scoring.

Software Engineering Co-op, Cognex Corporation

01/2023 – 08/2023

Natick, MA

- Developed a .NET wrapper layer to port an Edge Learning tool between two versions of VisionPro software.
- Delivered feature enhancements and bug fixes for VisionPro C++ APIs and the C# QuickBuild Edit Control UI.
- Expanded C++ and C# test coverage in the TeamCity CI infrastructure (API unit, BOM, color benchmark, and GUI sanity tests).
- Generated HTML and CHM documentation for .NET components with Doxygen and Sandcastle.
- Worked in an Agile scrum team.

EDUCATION

Northeastern University

09/2021 – 12/2023

M.S. in Electrical and Computer Engineering

Boston, MA

Beijing University of Technology

09/2017 – 05/2021

B.E. in Electrical Engineering

Beijing, China

PUBLICATIONS

* Equal contribution † Corresponding author

- Arash Akbari[†], Arman Akbari, Masih Eskandar, Qitao Tan, **Yixiao Chen**, Jingwu Luo, Bertha Pangaribuan, Liyun Zhang, Jennifer Dy, Geng Yuan, Xue Lin, Gaowen Liu, Stratis Ioannidis, Yanzhi Wang. **“ActQuant: Sub-4-bit Action-Guided Quantization for Vision-Language-Action Models.”** *NeurIPS 2026*. arXiv:2605.24011
- Ruining Yang^{*}, Muxing Wang^{*}, **Yixiao Chen**, Tongfei Guo, Yi Xu, Can Cui, Zichong Yang, Yitian Zhang, Ziran Wang, Yun Fu, Lili Su. **“Post-Training in End-to-End Autonomous Driving.”** *ECCV 2026 Workshop (Best Paper Award)*. arXiv:2607.08072

3. Amir Taherin, Sana Taghipour Anvari, Charles Amante, **Yixiao Chen**, Ruben Noroian, Zlatan Feric, Nicolas Bohm Agostini, Pu Zhao, José Cano, Bin Ren, Yanzhi Wang, David Kaeli. **“Hydra: Phase-Aware Workload Characterization of LLM Inference across Edge SoC Generations, Backends, and Quantization Levels.”** *IISWC 2026*. arXiv:2608.25053
4. Zhendong Mi, **Yixiao Chen**, Pu Zhao, Xiaodong Yu, Hao Wang, Yanzhi Wang, Shaoyi Huang[†]. **“Effective MoE-based LLM Compression by Exploiting Heterogeneous Inter-Group Experts Routing Frequency and Information Density.”** *ICML 2026*. arXiv:2602.09316
5. Arman Akbari[†], Ci Zhang, Arash Akbari, Lin Zhao, **Yixiao Chen**, Weiwei Chen, Xuan Zhang, Geng Yuan, Yanzhi Wang. **“Flash-WAM: Modality-Aware Distillation for World Action Models.”** *arXiv preprint, 2026*. arXiv:2606.05254
6. **Yixiao Chen**, Yanyue Xie, Ruining Yang, Wei Jiang, Wei Wang, Yong He, Yue Chen, Pu Zhao[†], Yanzhi Wang. **“Collaborative Compression for Large-Scale MoE Deployment on Edge.”** *ICASSP 2026*. arXiv:2509.25689
7. Xin Chen, Jia He, Maozheng Li, Dongliang Xu[†], Tianyu Wang, **Yixiao Chen**, Zhixin Lin, Yue Yao[†]. **“Are VLMs Ready for Lane Topology Awareness in Autonomous Driving?”** *ICASSP 2026*. arXiv:2509.16654
8. Timothy Rupprecht^{*†}, Pu Zhao^{*}, Amir Taherin^{*}, Arash Akbari^{*}, Arman Akbari^{*}, Yumei He^{*}, Tooba Imtiaz^{*}, Sean Duffy, Juyi Lin, **Yixiao Chen**, Rahul Chowdhury, Enfu Nan, Yixin Shen, Yifan Cao, Haochen Zeng, Weiwei Chen, Geng Yuan, Jennifer Dy, Sarah Ostadabbas, Xuan Zhang, David Kaeli, Edmund Yeh, Yanzhi Wang[†]. **“Human Cognition in Machines: A Unified Perspective of World Models.”** *arXiv preprint, 2026*. arXiv:2604.16592
9. **Yixiao Chen**^{*}, Ruining Yang^{*}, Xin Chen, Jia He, Dongliang Xu[†], Yue Yao[†]. **“From Static to Dynamic: a Survey of Topology-Aware Perception in Autonomous Driving.”** *ICCV 2025 Workshop (WDFM-AD)*. arXiv:2509.23641
10. Ruining Yang, Yi Xu, **Yixiao Chen**, Yun Fu, Lili Su. **“A Trajectory Generator for High-Density Traffic and Diverse Agent-Interaction Scenarios.”** *arXiv preprint, 2025*. arXiv:2510.02627
11. **Yixiao Chen**, Yue Yao[†], Ruining Yang, Md Zakir Hossain, Ashu Gupta, Tom Gedeon. **“Unsupervised Search for Ethnic Minorities' Disc-Cup Segmentation Training Set.”** *ICASSP 2025*. arXiv:2501.02442

AWARDS

Best Paper Award, ECCV 2026 Workshop for “Post-Training in End-to-End Autonomous Driving” 2026

SKILLS

Languages	Python, C/C++, C#, .NET (Core), SQL
ML Frameworks & Deployment	PyTorch, TensorRT, CoreML, llama.cpp, KTransformers, Vulkan, HIP/ROCm
Model Optimization	LLM quantization and pruning, MoE compression, model deployment and inference acceleration on edge devices
Models	DeepSeek V2/V3, Qwen 2/2.5, LLaMA 2, ViT, ViM, VLA / World Action Models
Tools	Linux servers, Git/Bitbucket (Stash), TeamCity CI, Doxygen, Sandcastle, Agile/Scrum